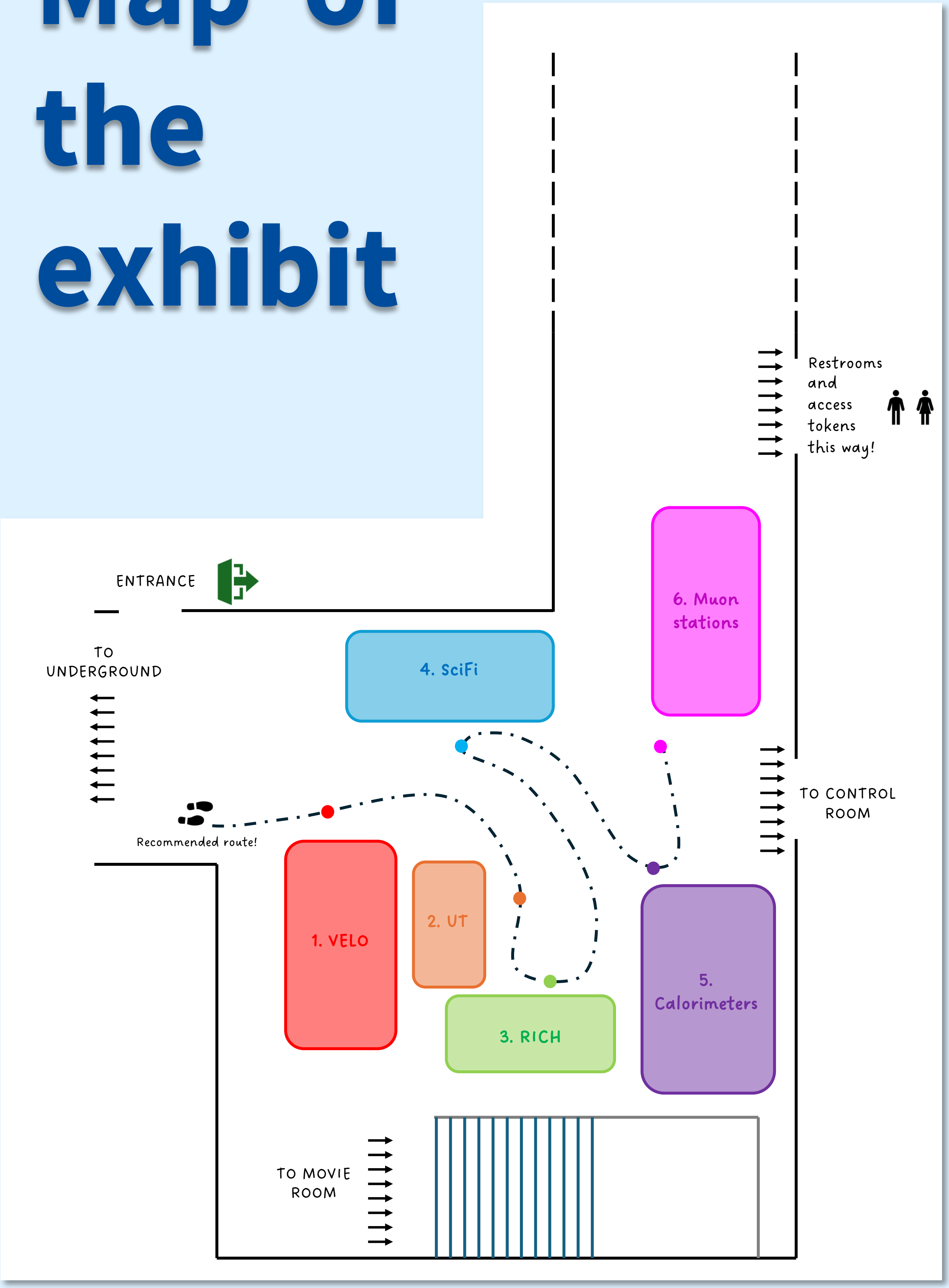
LHCb Guide for Guides!

July 2025

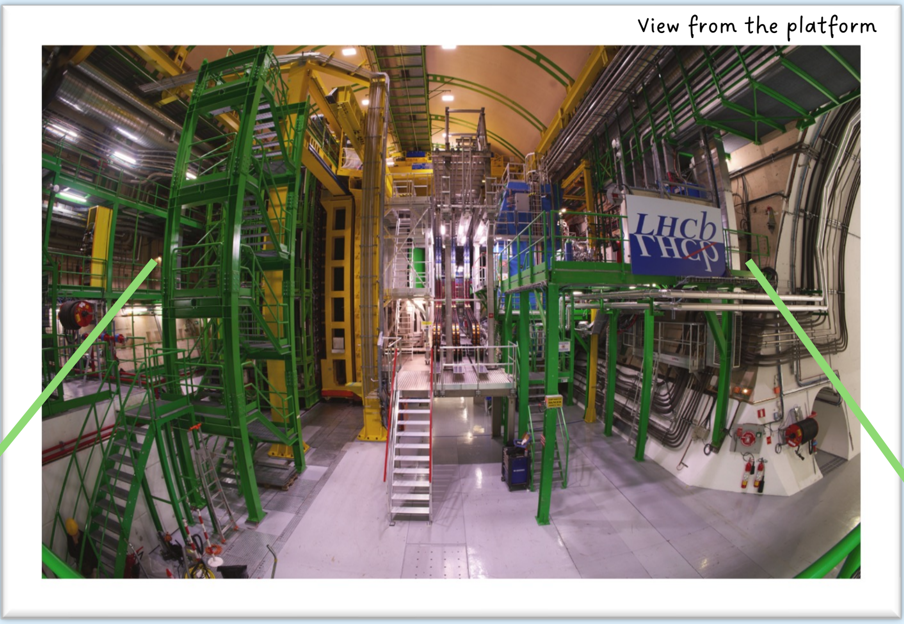
Map of the exhibit

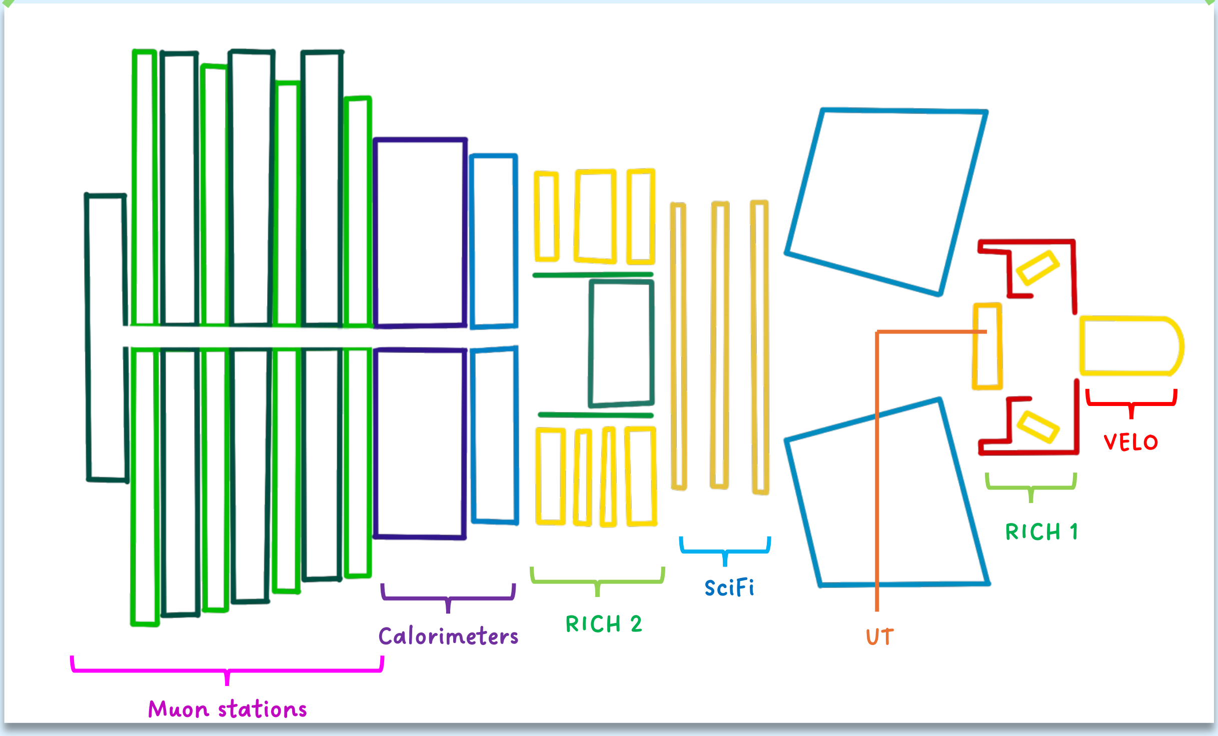


Introduction to LHCb

- LHCb is an experiment set up to explore what happened after the Big Bang that allowed matter to survive and build the Universe we inhabit today.
- Fourteen billion years ago, the Universe began with a bang. Crammed within an infinitely small space, energy coalesced to form equal quantities of matter and antimatter.
- But as the Universe cooled and expanded, its composition changed.
- Just one second after the Big Bang, antimatter had all but disappeared, leaving matter to form everything that we see around us — from the stars and galaxies, to the Earth and all life that it supports
- What happened to the anti-matter? Can study this by studying CP-violation (C = swap charges , P = mirror image)
- Can also study exotic states of quarks e.g. tetra- and pentaquarks
- Study of very rare processes that may be affected by new, undiscovered particles

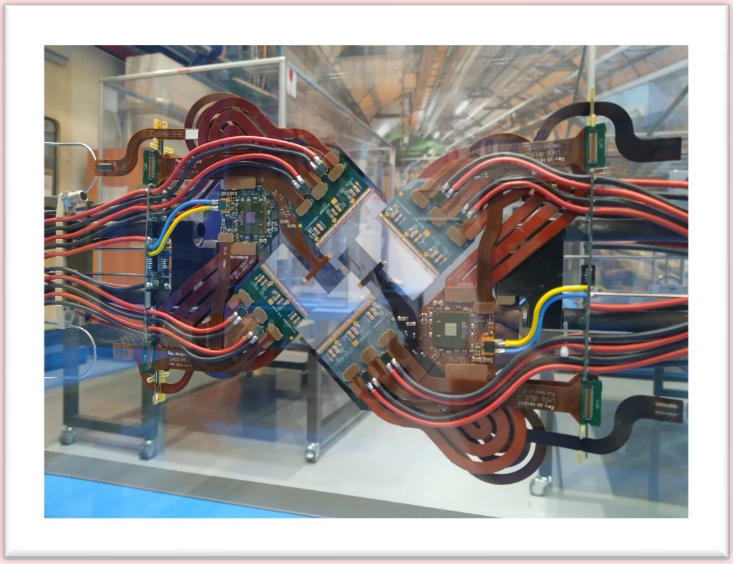
Looking at LHCb...





The VELO

Vertex **Lo**cator



WHAT DOES IT DO?

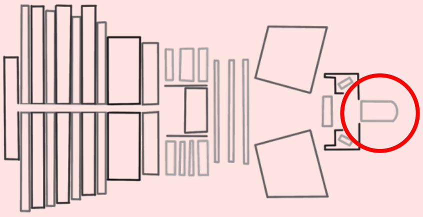
- Traces particle paths from the proton-proton collisions
- Allows us to reconstruct the **primary vertices** – where the b-hadrons originate

IN NUMBERS

- 52 modules
- 5.1 mm from the beam
- Pixels 55μm x 55μm
- Weighs 800 kg
- Temperature -40°C

HOW DOES IT DO IT?

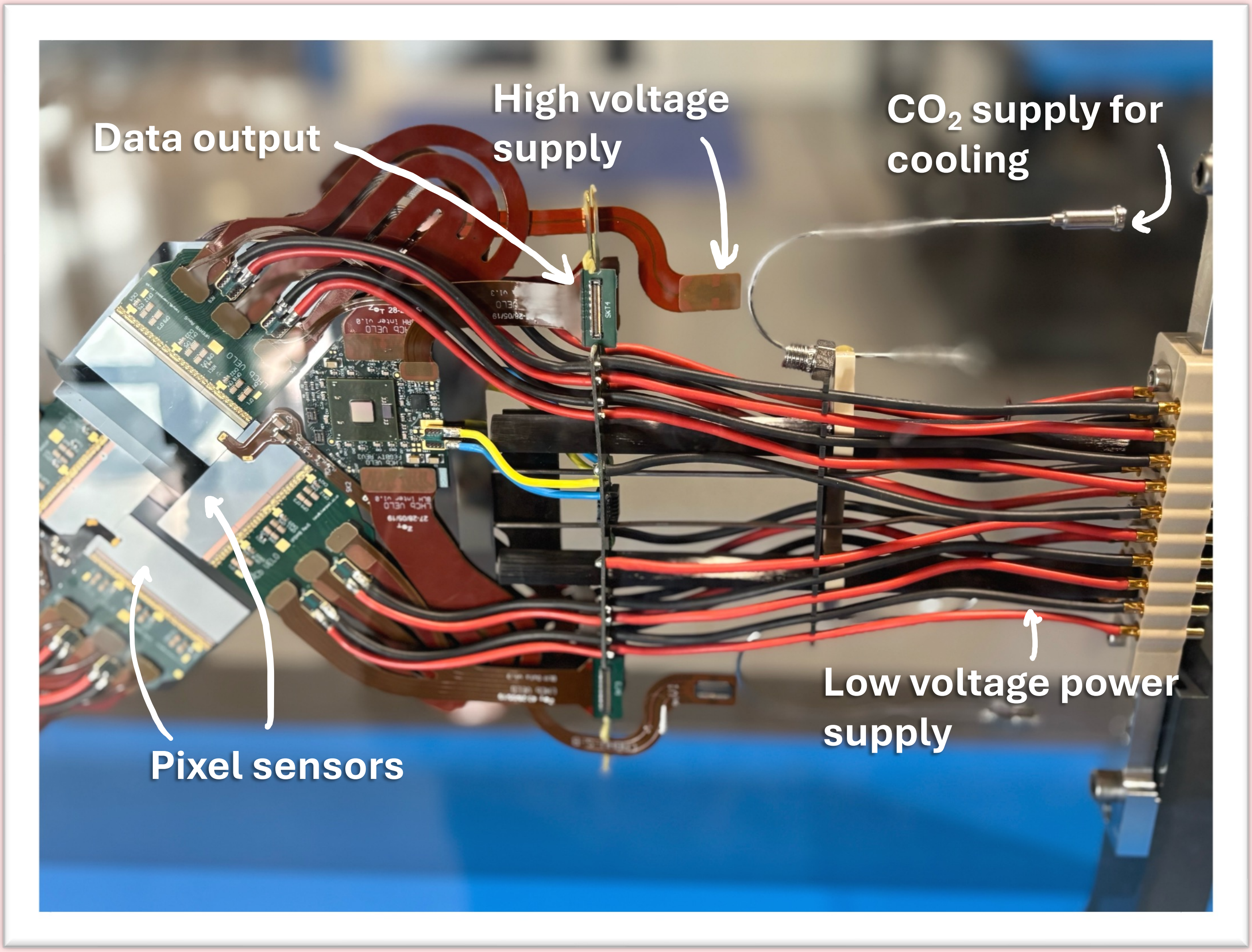
- Built from **silicon pixel sensors** (pixels 55μm x 55μm)
- **Measures the passage of charged particles**
- Sensors can be **moved towards** and **retracted from** the beam to get as close to the collisions without getting damaged
- Cooled by **liquid carbon dioxide** to withstand radiation

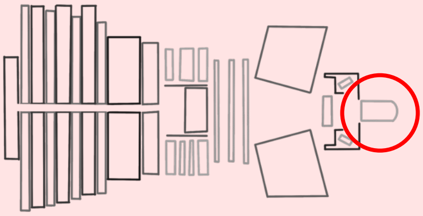


The VELO

Vertex **Lo**cator

In the exhibition…





The UT

Upstream **T**racker

WHAT DOES IT DO?

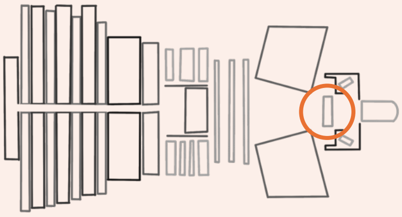
- Measures particle track positions **between the RICH1 and the magnet**
- Provides the **first estimate of track momentum**
- Important for particles that **decay after the VELO**
- Improves **track reconstruction and reduces fake track rate** by helping match tracks

HOW DOES IT DO IT?

- Built from **silicon microstrip sensors**
- Measures the **passage of charged particles** using semiconductors
- Has a **circular cut out** in the centre for the beam pipe to pass through

IN NUMBERS

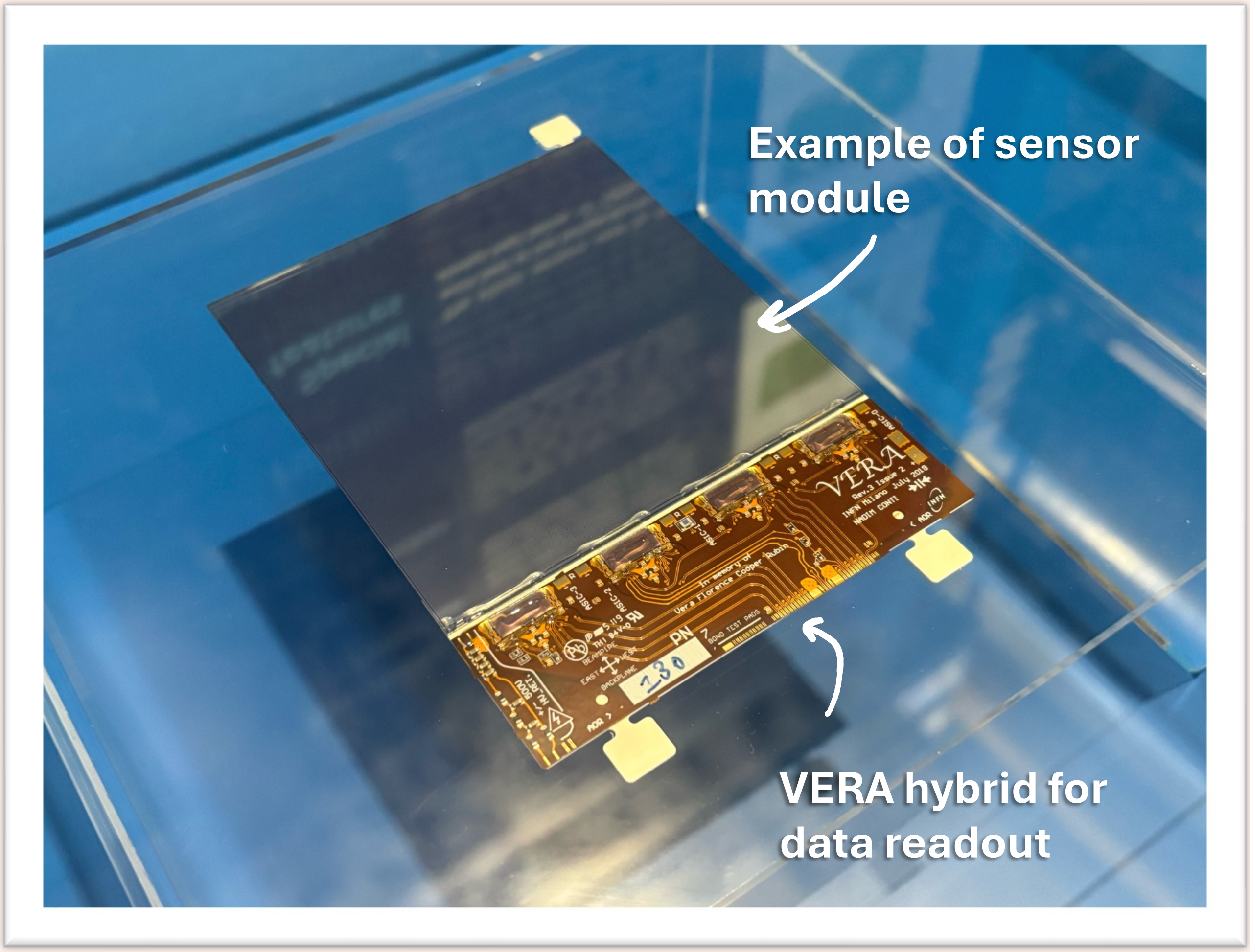
- **Four different** sensor designs
- Outer parts: each module has 512 strips, 187.5 μm wide
- Inner regions, each module has 1024 strips, 93.5 μm wide
- **968 modules in total**

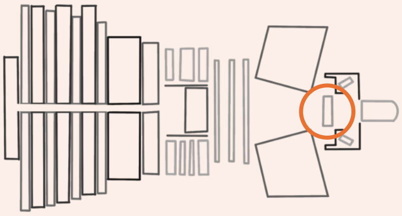


The UT

Upstream **T**racker

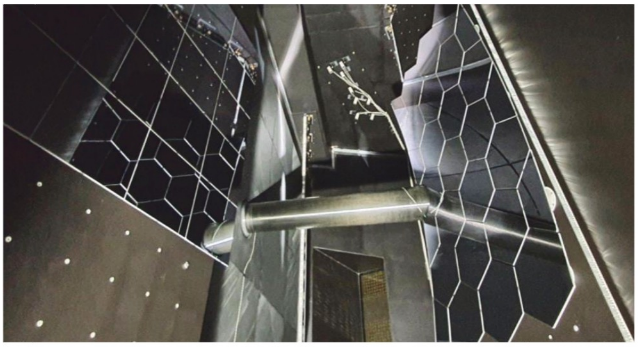
In the exhibition…





The RICH Detectors

Ring **I**maging **Ch**erenkov

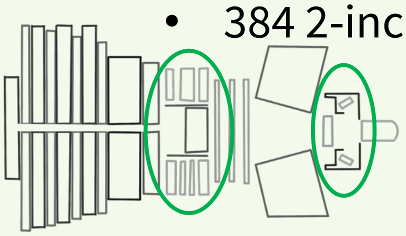


WHAT DO THEY DO?

- Provide **particle identification** information
- Particles that travel faster than light in the radiation medium produce **Cherenkov radiation** – cones of light
- Knowing **angle of light** cone and **momentum** allows mass measurement – can identify particle

IN NUMBERS

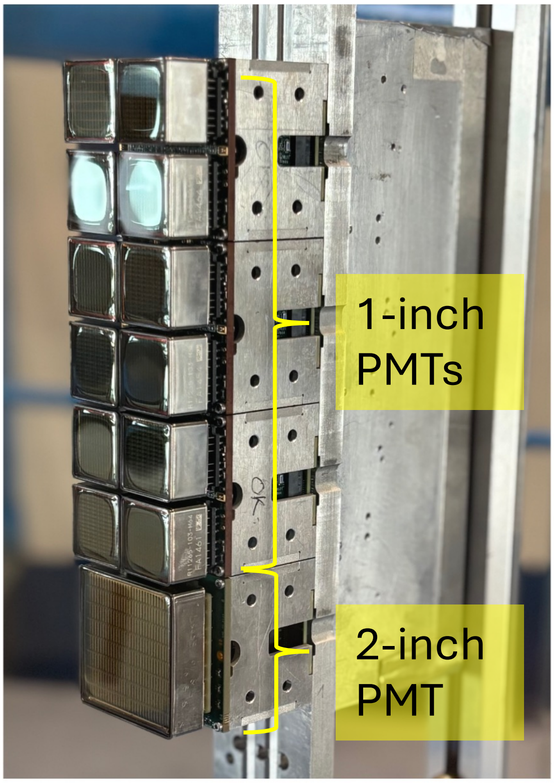
- 2 detectors
- RICH1
    - 1888 1-inch MaPMTs
    - 4 spherical mirrors (each 2.9 kg)
    - 16 planar mirrors (each 3.3 kg)
- RICH2
    - 768 1-inch MaPMTs
    - 384 2-inch MaPMTs



HOW DO THEY DO IT?

- Using **focusing mirrors** and the **photon detectors** to measure Cherenkov photons
- Use devices known as multi-anode photomultiplier tubes (MaPMTs)

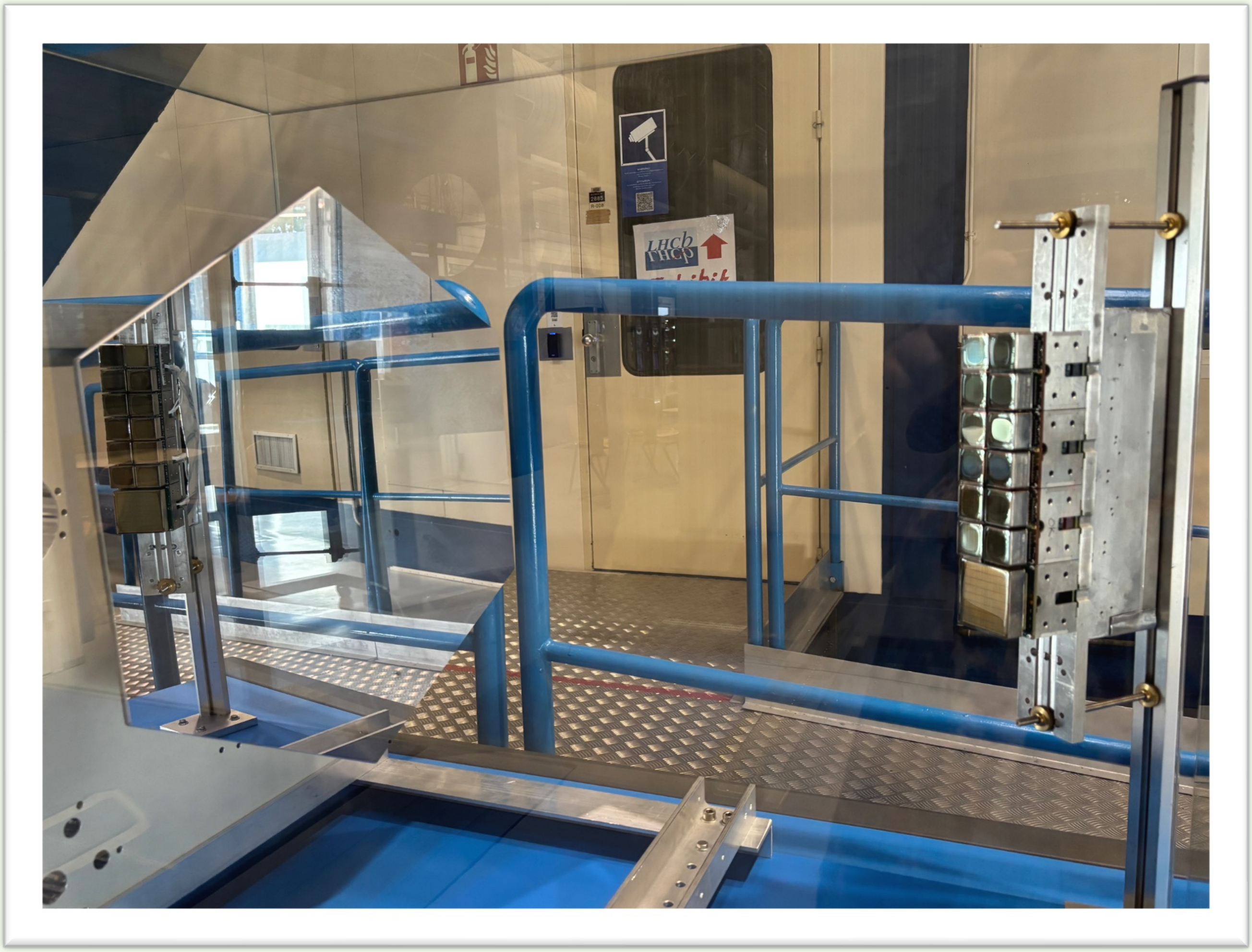
In the exhibition…



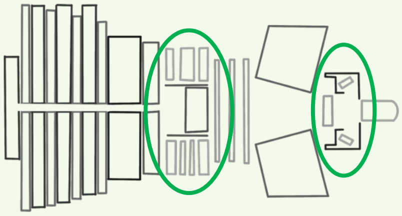
The RICH Detectors

<u>**R**</u>ing <u>**I**</u>maging <u>**Ch**</u>erenkov

<u>In the exhibition…</u>



PMTs and a large mirror



The SciFi

Scintillating **Fi**bre

WHAT DOES IT DO?

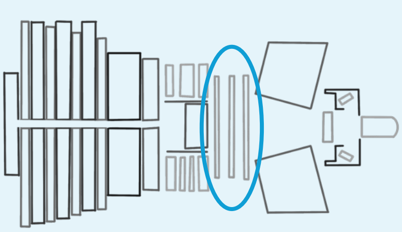
- Measures the position of a charged particle
- Detector hits used to reconstruct particle trajectories and their momenta

HOW DOES IT DO IT?

- Charged particle passing through the fibres in the detector will emit light
- Light travels along the fibres and collected by photomultipliers
- Calculate position of the particle from the light signals

IN NUMBERS

- Total height: 5 m
- 12 detection planes in 3 stations
- 11,000 km of fibres that have 250 µm diameter
- More than 500,000 silicon photomultiplier channels used to detect the light

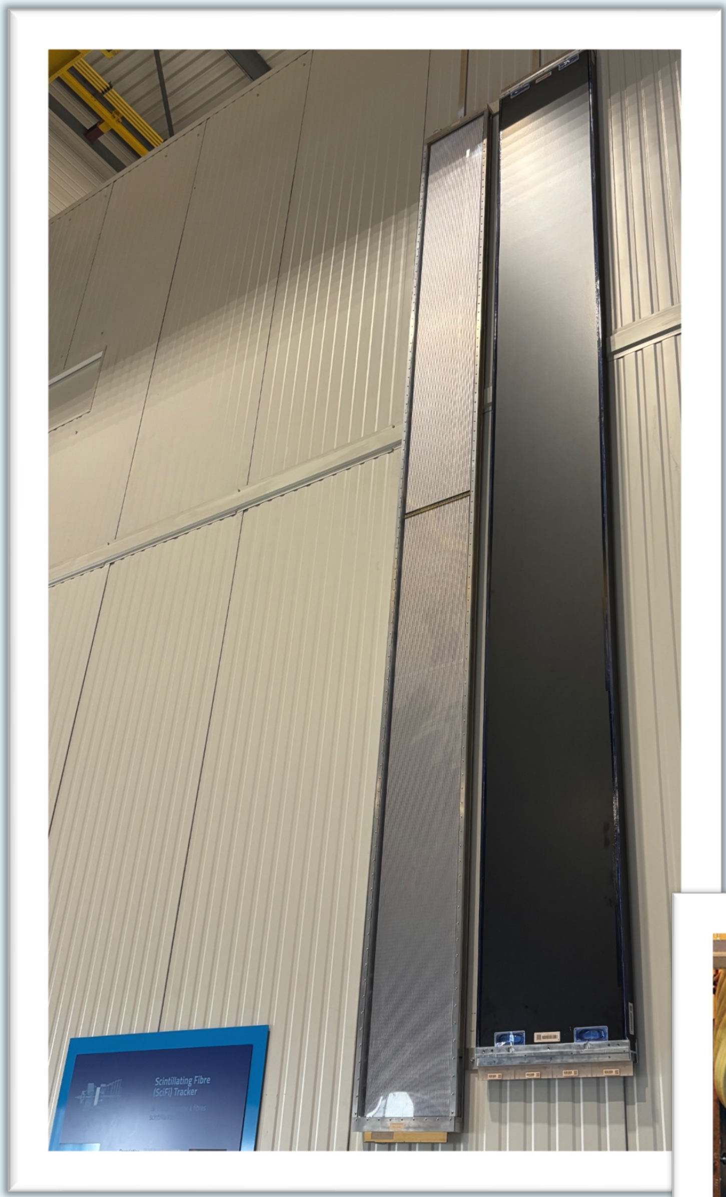


The SciFi

Scintillating **Fi**bre

In the exhibition…

An example module



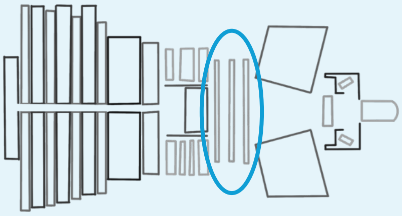
Magnet

Charged particles bend in magnetic field – positive and negative particles bend opposite ways

LHCb magnet is “warm” – not superconducting. Field is 4 Tm (typical fridge magnet is 0.005 T)

Made of two coils – 27 tons each
- 7.5 m long
- 4.6 m wide
- 2.5 m high

Supported by steel yoke – 1450 tons



The Calorimeters

WHAT DO THEY DO?

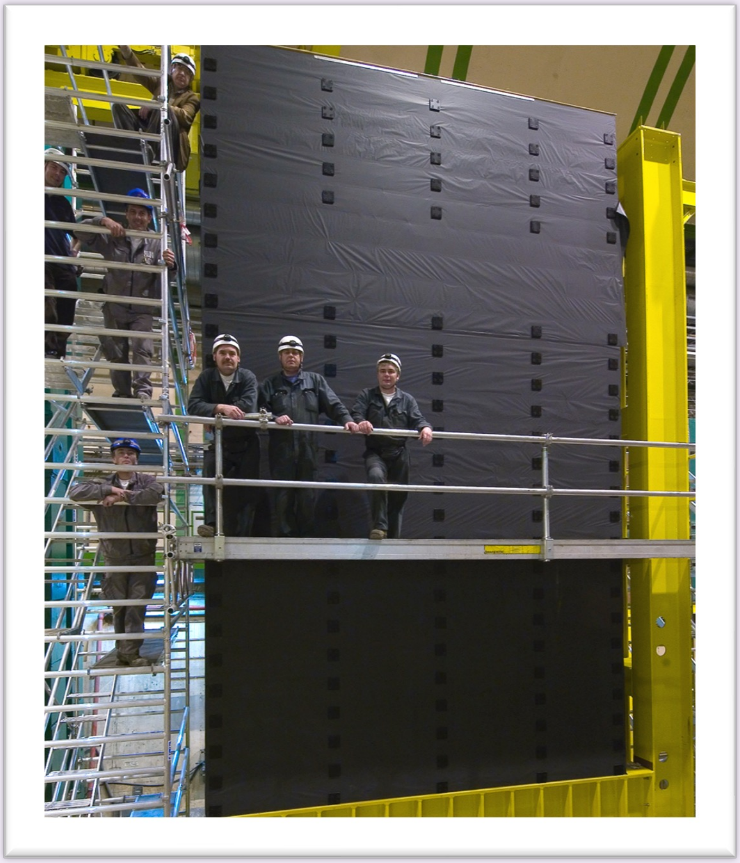
- Measure the final energies of particles
- Two types:
  - Hadronic (HCAL)
  - Electromagnetic (ECAL)

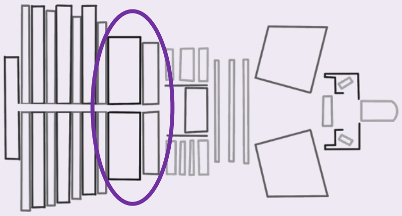
HOW DO THEY DO IT?

- Designed so that particles entering the detector will interact and produce 'showers' of new particles
- Layers of different materials to stop the particles and to measure the light from the shower particles

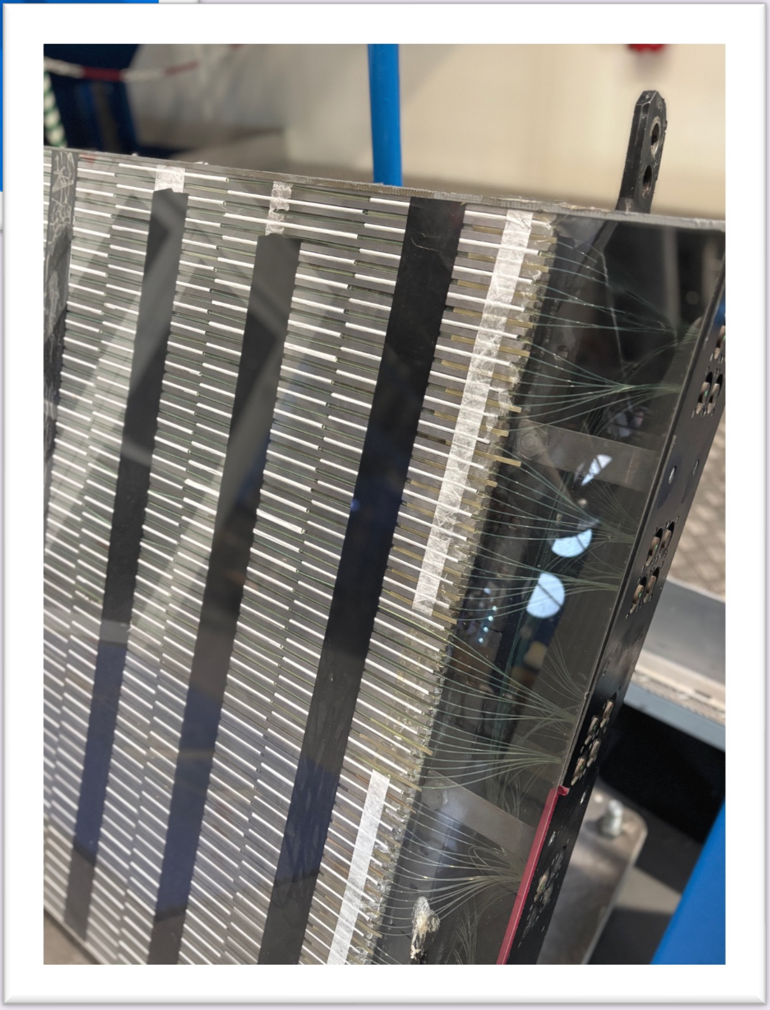
IN NUMBERS

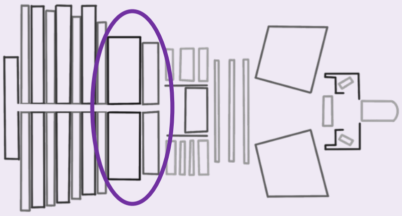
- 6016 ECAL cells, each have
  - 66 layers of lead (2 mm thick)
  - 67 layers of scintillator (4 mm thick)
- 1488 HCAL cells, with staggered iron and plastic scintillator



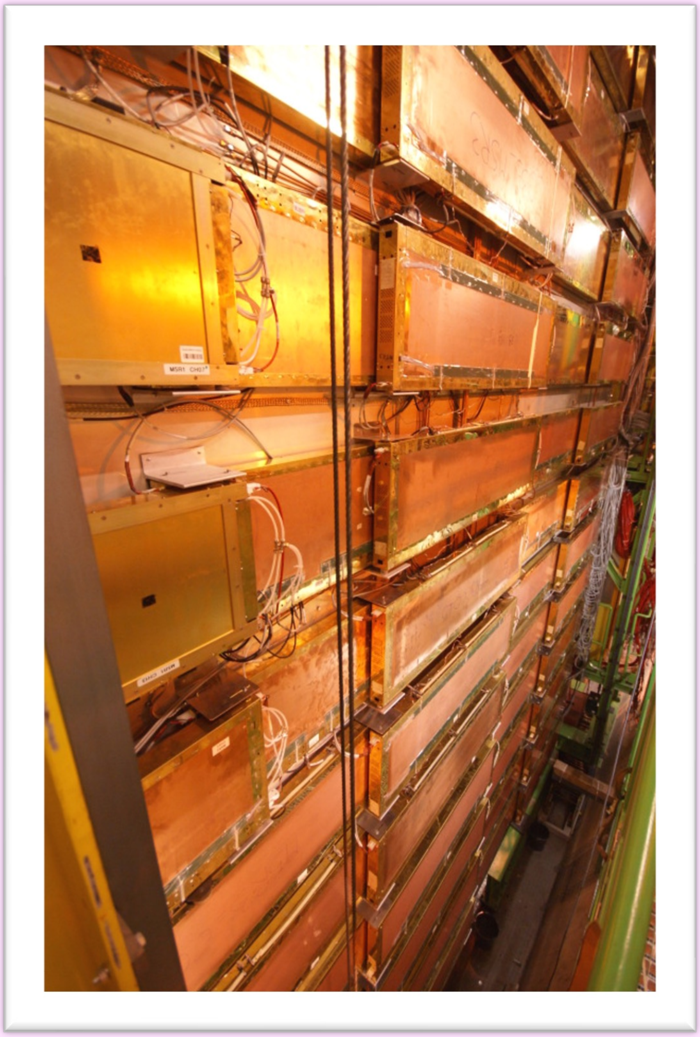


The Calorimeters





Muon stations



WHAT DO THEY DO?

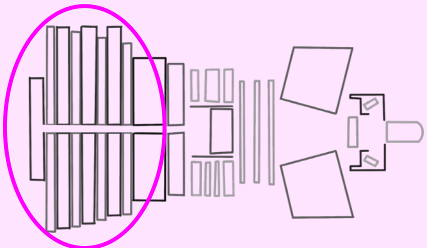
- Muons travel through the calorimeters and are stable within LHCb
- Detected at the very end by dedicated muon stations that identify muon tracks

HOW DO THEY DO IT?

- Uses multiwire proportional chambers (MWPC) which contain gases
- Muons ionize the gases in the chambers and this ionization produces an electrical signal recorded by wires inside the chambers

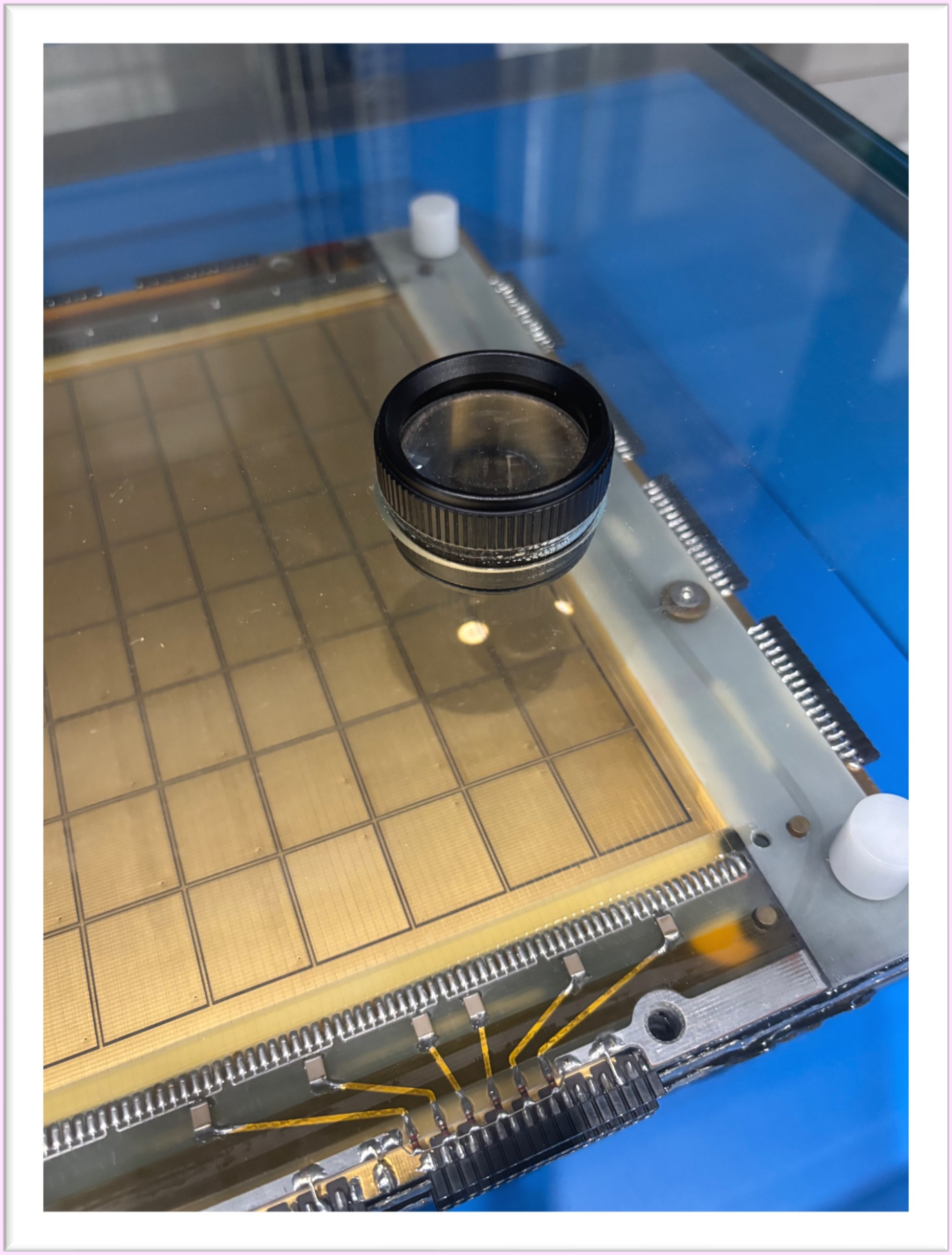
IN NUMBERS

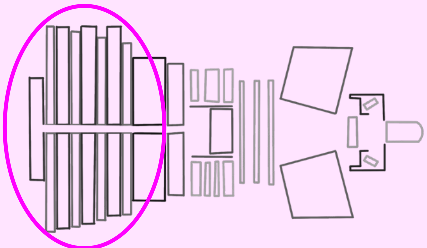
- 4 stations in total
- Total of 1104 MWPCs
- Total area covered is 385 m^2
- More than 1200 km of wire



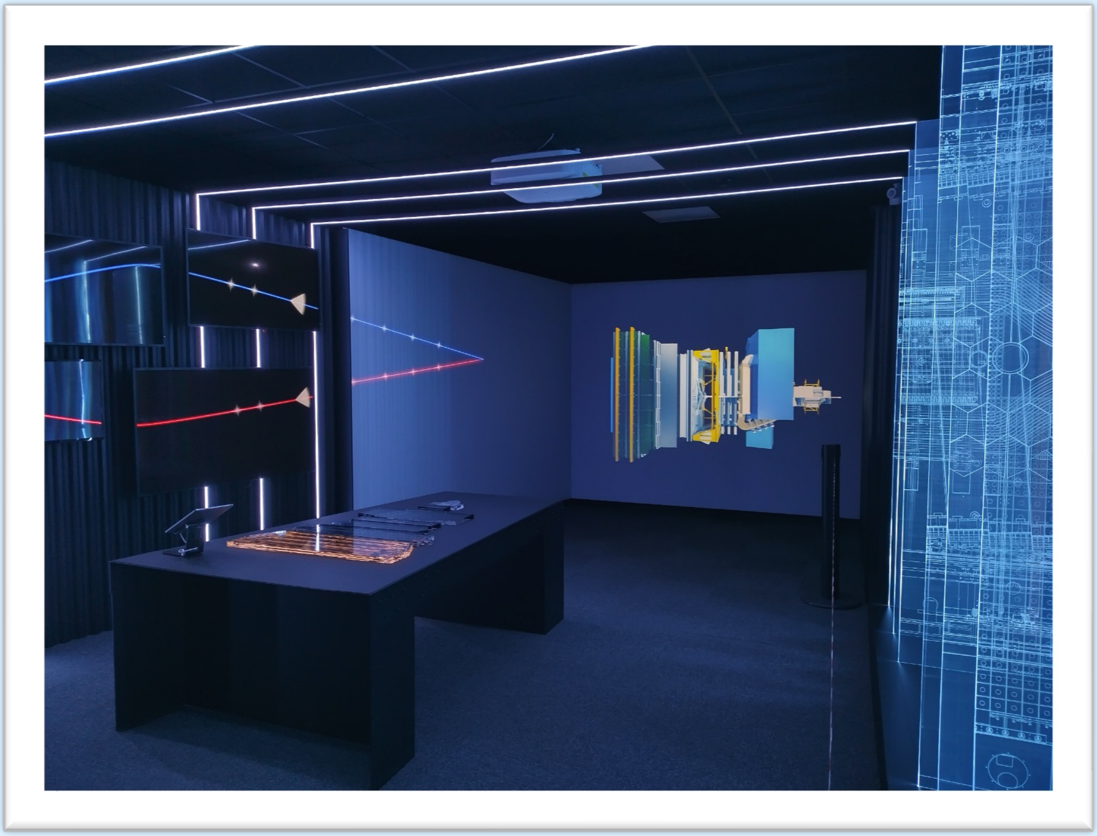
Muon stations

In the exhibition...
An example module with magnifying glass





The cinema room

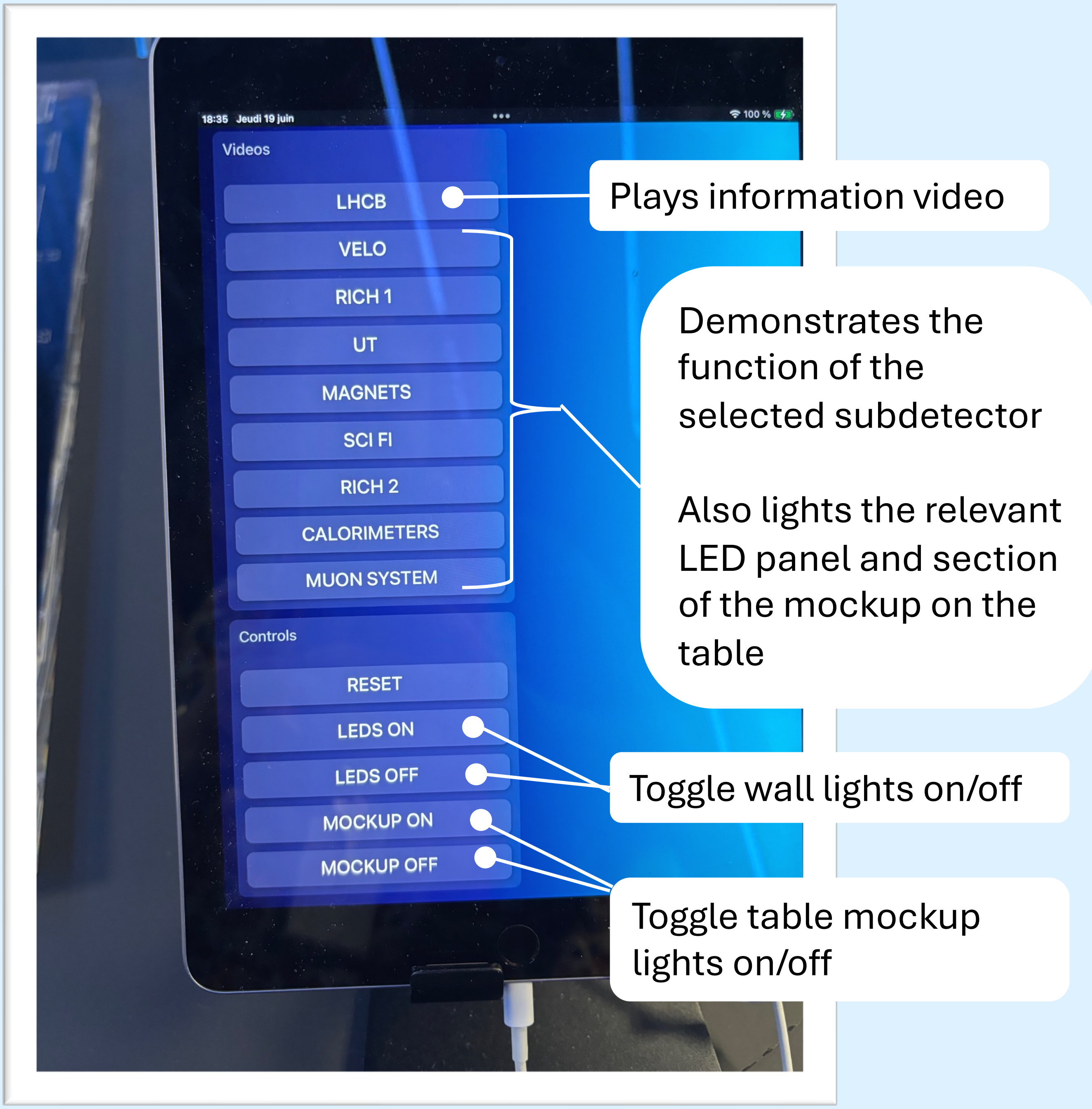


The iPad on the desk controls all lights and the videos.

There is a video that explains the physics behind LHCb, and then a button that shows how particle tracks are measured in each subdetector.

The main light switch is the white button on your left as you walk in the door.

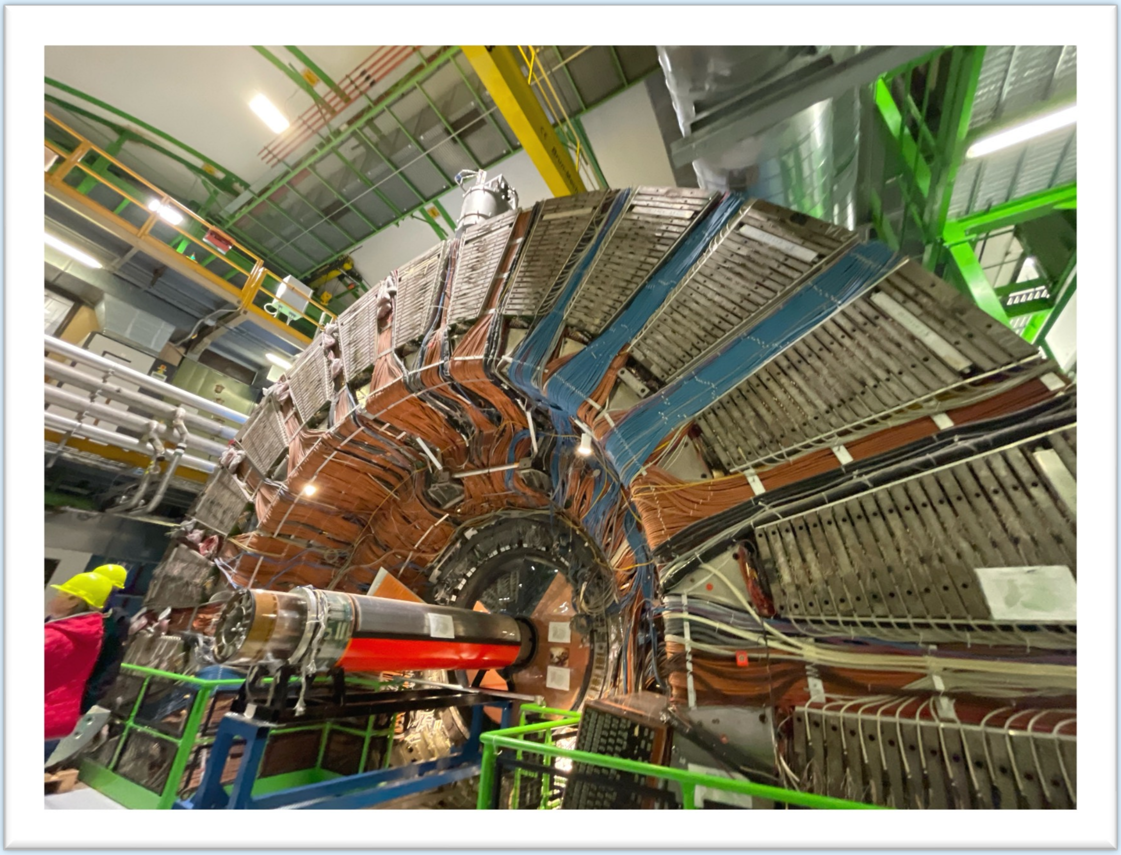
The cinema room



Going underground?

- Helmets found by the doors to the elevators
- Tokens can be found beyond the door to the left of the muon station exhibit (see map on page 3)
- LHCb deepest of all LHC experiments at 103 m!

DELPHI



- Before the LHC had LEP (Large Electron Positron Collider) which had 4 collision points
- Operated 1989 – 2000
- LEP energy ~100 times less than LHC energy
- Where LHCb lives now, used to be DELPHI (Detector with Lepton, Photon and Hadron Identification)
- Barrel weighs 3200 T and is 11 m tall
- Cylindrical layers: vertex detector, tracking chambers, RICH detectors, solenoid magnet, calorimeters, muon detectors

LHCb viewing platform

- Two levels
  - Upper: 1 m above beamline
  - Lower: 1.5 m below beamline
- LHCb shaft is 103 m deep and 10 m in diameter